

Software Design Diagram Generator

IT314 Software Engineering

Group 27 · Concept Poster · Sprint 1

From plain-language requirements to reviewed, traceable UML and DFD diagrams — in minutes, not days

Why this tool

- Writing design diagrams from a requirements document is slow and error-prone.
- Ambiguous requirements slip through into the design.
- Diagrams drift away from the requirements they came from.

Who it is for

- Developers and student teams
- Software architects / designers
- Project managers and team leads
- System administrators

English input · desktop Chrome, Firefox and Edge

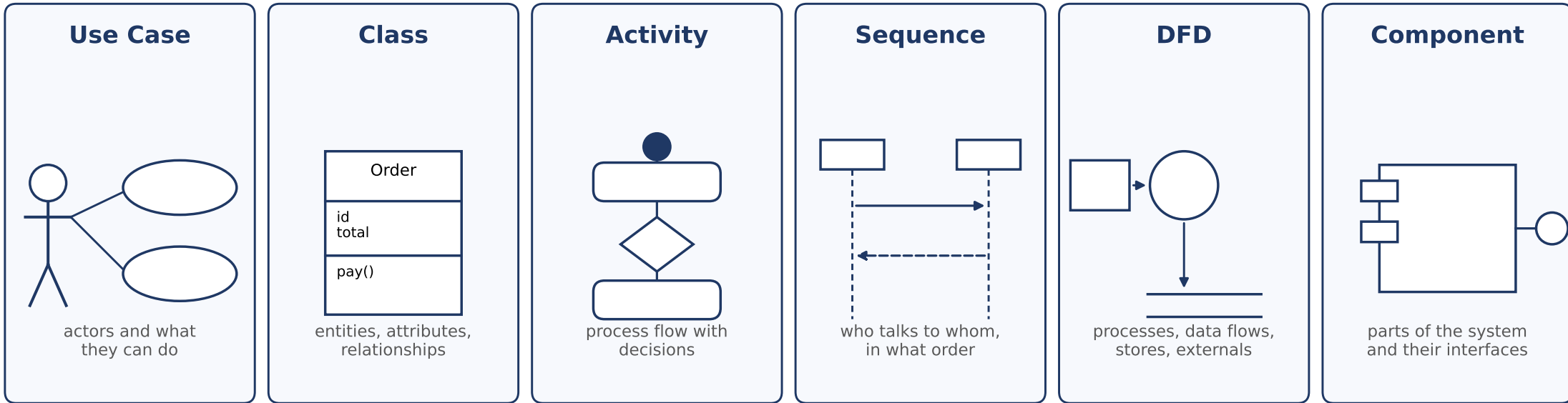
What sets it apart

- Every element traces back to the requirement it came from
- Ambiguity flagged and explained
- Assumptions clearly marked
- Editable, versioned diagrams
- Roles for team collaboration
- PM report and admin dashboard

How it works — the pipeline

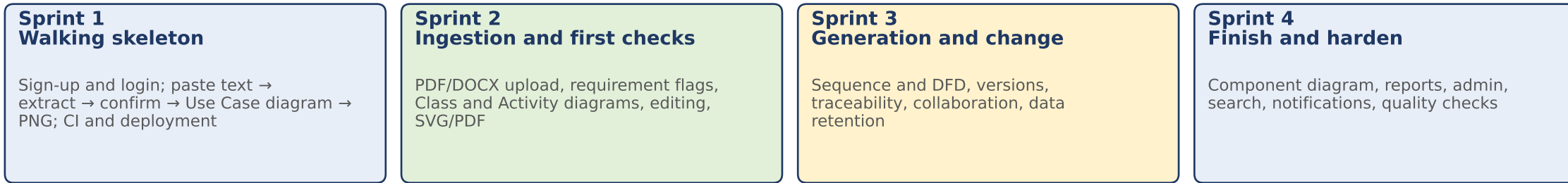


Six diagram types, each following its own notation



Elements without requirement backing are marked as assumptions, so the diagram never presents a guess as fact.

Roadmap — 155 user stories in 4 sprints



Built with: Vercel (frontend) → Render (backend) → database + encrypted file storage ↔ Google Gemini (AI service) · GitHub + Slack · Agile / Scrum, 4 sprints

Quality targets: generation ≤ 120 s (median ≤ 60 s) · 5 concurrent users · HTTPS and encryption at rest · 30-day inactivity deletion · latest two versions of Chrome, Firefox, Edge